Problem 10 | Problems T4_PhaseDiagrams I.pdf #10 | Cu-Sn · eutectoids

Reactions and Eutectoids in the Cu-Sn System

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 4

10. Considering the Cu-Sn phase diagram: a. Indicate the type of transformations, and write the reactions that take place at T_1 = 798 °C; T_2 = 755 °C; T_3 = 588 °C; T_4 = 415 °C; T_5 = 1084.87 °C. b. For a Cu-20%Sn alloy, indicate the phases and microconstituents, their composition and amount at T_1: 521 °C and T_2: 519 °C. c. Determine the percentage of Sn in an alloy having 54% liquid phase at 900 °C. Is this alloy in solid state at 700 °C?

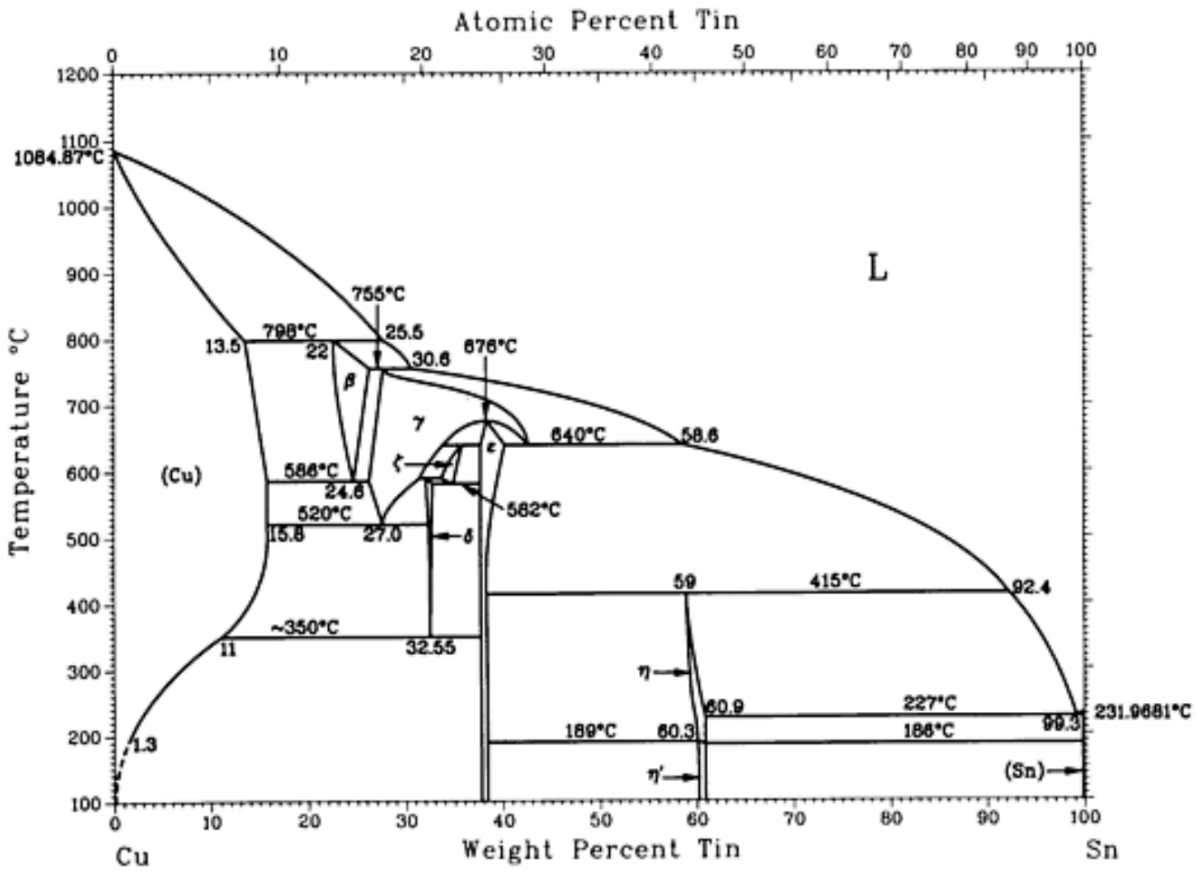


Figure 17. Cu-Sn phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 4).

Official Solution: a) peritectic; peritectic, eutectoid; peritectic and solidification of Cu. b) 521 °C: $C_{(Cu)}$ = 15.8%Sn; C_γ = 27%Sn; X_γ = 0.37; $X_{(Cu)}$ = 0.63; X(eutectoid): 0.48; X(proeutectoid Cu): 0.52. 519 °C: $C_{(Cu)}$ = 15.8%Sn; C_δ = 32%Sn; X_δ = 0.26; $X_{(Cu)}$ = 0.74; X(eutectoid): 0.37 and X(proeutectoid Cu): 0.63. c) 14.5%Sn

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Cu-Sn diagram of the sheet (wt% Sn). Printed values: 798 °C with 13.5, 22 and 25.5 wt% Sn; 755 °C with 30.6; 586 °C with 24.6; 520 °C with 15.8 and 27.0; 415 °C with 59 and 92.4. The rest are **readings of the figure** (±1 wt%).
2. The statement's 588 °C is the 586 °C line of the diagram.
3. The type of each reaction follows from its shape [Session 8 Slide 6].

Data

Line	Phases and compositions (wt% Sn)
798 °C	(Cu) 13.5, β 22, L 25.5
755 °C	β ≈ 24, γ ≈ 25.5, L 30.6
586 °C	(Cu) 15.8, β 24.6, γ ≈ 25.5
520 °C	(Cu) 15.8, γ 27.0, δ ≈ 32
415 °C	ε ≈ 38, η 59, L 92.4
900 °C	(Cu) ≈ 9, L ≈ 19 (on the solidus and liquidus)

PHASE 2 · FRAMES & STRATEGY

Frame: the Cu-Sn diagram (wt% Sn, °C); all the work is on horizontal lines, so no change of basis is needed.

Strategy. (a) Name each line by its shape. (b) Lever rule on the tie lines just above and just below 520 °C; the microconstituents come from the history of the alloy (it crosses 798, 586 and 520 °C). (c) Write the lever rule for the liquid at 900 °C and solve it for C_0.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Transformations

T	Reaction on cooling	Type
T_1 = 798 °C	$(\mathrm{Cu})\,(13.5) + L\,(25.5) \rightarrow \beta\,(22)$	peritectic
T_2 = 755 °C	$\beta\,(\approx 24) + L\,(30.6) \rightarrow \gamma\,(\approx 25.5)$	peritectic
T_3 = 586 °C	$\beta\,(24.6) \rightarrow (\mathrm{Cu})\,(15.8) + \gamma\,(\approx 25.5)$	eutectoid
T_4 = 415 °C	$\varepsilon\,(\approx 38) + L\,(92.4) \rightarrow \eta\,(59)$	peritectic
T_5 = 1084.87 °C	$L \rightarrow (\mathrm{Cu})$ for pure Cu	solidification of a pure metal

b) Cu-20 wt% Sn History. $13.5 < 20 < 22$: at 798 °C the peritectic consumes all the liquid and leaves (Cu) + β. On cooling, β is transformed at 586 °C by the eutectoid into (Cu) + γ, and γ at 520 °C by a second eutectoid into (Cu) + δ.

Just above 586 °C ((Cu) 15.8, β 24.6; length 8.8):

$$W_{\beta} = \frac{20 - 15.8}{24.6 - 15.8} = \frac{4.2}{8.8} = 0.477 \qquad W_{(\mathrm{Cu})} = \frac{24.6 - 20}{8.8} = \frac{4.6}{8.8} = 0.523$$

This (Cu) is **proeutectoid**, and this β becomes the **eutectoid** microconstituent at 586 °C.

At 521 °C (phases), (Cu) 15.8 and γ 27.0 (length 11.2):

$$W_{\gamma} = \frac{20 - 15.8}{27.0 - 15.8} = \frac{4.2}{11.2} = 0.375 \qquad W_{(\mathrm{Cu})} = \frac{27.0 - 20}{11.2} = \frac{7.0}{11.2} = 0.625$$

Microconstituents at 521 °C: proeutectoid (Cu) 0.523 and eutectoid [(Cu) + γ] 0.477.

At 519 °C (phases), below the 520 °C eutectoid $\gamma\,(27.0) \rightarrow (\mathrm{Cu})\,(15.8) + \delta\,(32)$; (Cu) 15.8 and δ ≈ 32 (length 16.2):

$$W_{\delta} = \frac{20 - 15.8}{32 - 15.8} = \frac{4.2}{16.2} = 0.259 \qquad W_{(\mathrm{Cu})} = \frac{32 - 20}{16.2} = \frac{12}{16.2} = 0.741$$

Microconstituents at 519 °C, counted with respect to the 520 °C reaction: the γ present at 521 °C (0.375) becomes eutectoid [(Cu) + δ], and all the (Cu) that existed before (0.625) counts as proeutectoid. That (Cu) is made of the proeutectoid (Cu) of 586 °C (0.523) plus the (Cu) of the first eutectoid ($0.625 - 0.523 = 0.102$).

c) Alloy with 54 % liquid at 900 °C At 900 °C the tie line goes from (Cu) at $C_S \approx 9$ to L at $C_L \approx 19$ wt% Sn (readings). The lever rule for the liquid:

$$W_L = \frac{C_0 - C_S}{C_L - C_S} \;\Rightarrow\; C_0 = C_S + W_L\,(C_L - C_S) = 9 + 0.54 \times (19 - 9) = 9 + 5.4 = 14.4\ \mathrm{wt\%\ Sn}$$

Solid at 700 °C? $13.5 < 14.4 < 22$, so at 798 °C the peritectic uses up all the liquid: below 798 °C the alloy is (Cu) + β, both solid. At 700 °C the (Cu) boundary is at about 14.5 wt% Sn (it goes from 13.5 at 798 °C to 15.8 at 586 °C), so the alloy (14.4) is (Cu), practically single-phase. The key gives 14.5 wt% Sn; the 0.1 difference is well within the ±1 wt% of the readings and does not change the answer. **Yes, it is solid at 700 °C.**

PHASE 4 · CHECKS & INTERPRETATION

Checks

- 519 °C: $0.741 \times 15.8 + 0.259 \times 32 = 11.71 + 8.29 = 20.0$ wt% Sn.
- 521 °C: $0.625 \times 15.8 + 0.375 \times 27.0 = 9.88 + 10.13 = 20.0$ wt% Sn.
- (c): with $C_0 = 14.4$, $W_L = (14.4 - 9)/(19 - 9) = 0.54$.
- **Limiting cases** (fractions dimensionless, compositions in wt% Sn): in (c), $W_L \to 0$ as $C_0 \to C_S = 9$ and $W_L \to 1$ as $C_0 \to C_L = 19$; 14.4 lies a little past the middle of the tie line, consistent with 0.54 of liquid.

Interpretation The Cu-rich side of Cu-Sn (bronzes) shows how one alloy can cross several invariant reactions: a peritectic during solidification and two eutectoids in the solid. Each eutectoid turns a single solid phase into a fine two-phase mixture.

Check against the official solution

All values agree with the key: 0.37/0.63 and 0.48/0.52 at 521 °C; 0.26/0.74 and 0.37/0.63 at 519 °C (the key counts microconstituents with respect to the 520 °C reaction, as explained in part b); 14.5 % Sn. The key's part (c) does not answer the question about 700 °C.

> ERRATUM IN THE WORKED SOLUTION
> The annotated diagram of Solution problems Phase diagrams I.pdf, p. 35, labels the field above the liquidus at about 50 % Sn and 900 °C as "(Cu)+ε"; that field is the liquid L, as the sheet's own diagram shows.